

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Square A has side lengths that are **246** times the side lengths of square B. The area of square A is ***k*** times the area of square B. What is the value of ***k***?

Answer

- A. **60,516**
- B. **492**
- C. **246**
- D. **123**

Correct Answer: A

Rationale

Choice A is correct. The area of a square is ***s*²**, where ***s*** is the side length of the square. Therefore, the area of square B is ***b*²**, where ***b*** is the side length of square B. It's given that square A has side lengths that are **246** times the side lengths of square B. Therefore, the side length of square A can be represented by the expression **246*b***. It follows that the area of square A is **(246*b*)²**, or **60,516*b*²**. It's given that the area of square A is ***k*** times the area of square B, so **60,516*b*² = *k**b*²**. Therefore, the value of ***k*** is **60,516**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.